SECOND REVIEW · ACADEMIC PROTOTYPE · NOT AN OPERATIONAL SYSTEM

Component E — Network and coordination screen

How accounts connect, and why a high score is a screen, not an accusation.

This brief is for the teammate who presents the graph. The score is an unweighted average of three behaviour signals. It is not a verdict that an account is a bot or a campaign.

1. What this agent does

Component E builds a graph of public interactions: forwards, replies, and mentions. It finds communities, finds accounts that sit between communities, and computes a screening score for similar posting behaviour.

2. Where it sits

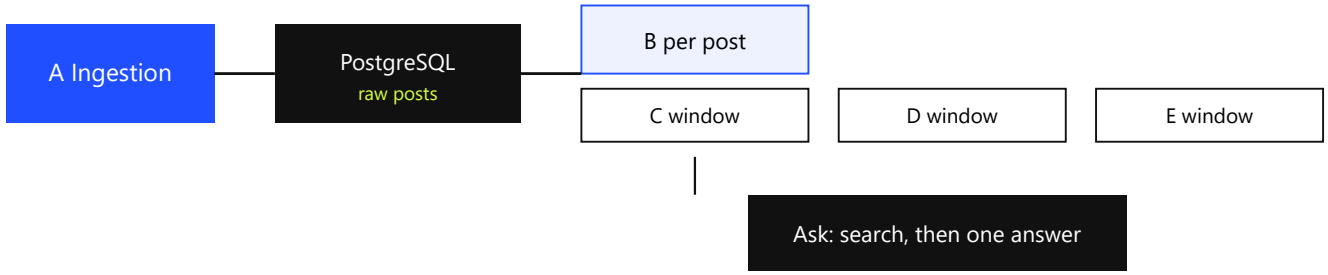


Figure 1. Posts land in Postgres. B reads one post at a time. C, D, and E recompute a time window. Ask searches the short summaries those agents write.

E recomputes a window. Edges are directed, because a forward has a direction. Community detection needs an undirected copy of that graph. The code makes a real copy, not a view, then runs Louvain. The import is `import community` from the package `python-louvain`.

3. The picture the page draws

On screen	Meaning
Node	A channel or a public author.
Line	A forward, reply, or mention.
Size	Betweenness: how often the account sits on paths between others.
Colour groups	Louvain communities on the undirected copy.
Red ring	The screening score is at least 0.70.

The API returns at most 500 nodes, the ones with the highest betweenness. Edges are only those whose both ends are in that set. That keeps the canvas from drawing dangling lines.

4. The three signals

Signal	Question it asks
Posting synchrony	Do this account's posts land within two minutes of other posts unusually often?
Age and volume	Is a young account posting a lot? Missing creation dates weaken this signal.
Duplicate content	Is the text unusually similar to other texts in the window? Similarity uses TF-IDF, not the neural embedder.

The score is `(s1 + s2 + s3) / 3`. There are no learned weights. A flag is raised at 0.70. Both choices are locked so the review can be explained on a whiteboard.



Figure 2. Three equal signals. No signal is allowed to outweigh the others by a trained weight.

5. What to show

Open **Network**. Name one community count and one connector. If the flagged count is zero, say the screen found no account over 0.70 in this window. Zero is a result, not a broken page.

> **One sentence.** "A red ring means the three behaviour scores average at least 0.70. It does not name an operator."

6. Related parts

Part	Relation to E
Tables	`network_nodes_e`, `network_edges_e`
API	`GET /api/network/graph`
Ask	May describe a community summary. It must not upgrade a screen into an attribution.
Phase 5	An offline coordination file was planned to score two of the three signals. That benchmark has not been run. Do not quote precision or recall.

7. Done, and not done

Done	Not claimed
Directed graph, undirected Louvain, three-signal average, 500-node cap, Network page.	No bot label, no legal finding, no published F1. The offline benchmark is still future work.

> **Do not say** "we detected a campaign" or "this account is fake".

8. Likely questions

Question	Answer
Why undirected Louvain?	The standard Louvain implementation we use expects an undirected graph. We keep the directed edges for the drawing.
Why equal weights?	A learned weight would need a labelled set we have not evaluated. An equal average is explainable.
Why is the same channel listed twice sometimes?	An older run could leave a second snapshot. The page is supposed to keep the latest window. If a duplicate remains, it is a snapshot bug, not two actors.